

Jeonggyu Hwang

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EXPERIENCE

POSTECH

Pohang-si, Gyeongsangbuk-do, South Korea

Pohang University of Science and Technology

Research Participating Undergraduate Student

February 2024

Department of Chemical Engineering

- FIMDEM Lab. - Functional Inorganic Materials Design & Manufacturing
- PI: Prof. Son, Jaesung Ph.D.
- Research topic: Solution-Processed Thin Film, Thermoelectric Device Material

Korea University

Seongbuk-gu, Seoul, South Korea

Research Participating Undergraduate Student

July 2023 - August 2023

College of Engineering, Department of Integrative Energy Engineering

Graduate School of Energy and Environment

- Electrochemistry Lab. - Solarrino
- PI: Prof. Jun, Yongseok Ph.D.
- Research topic: Perovskite Solar Cells

INDUSTRY

Philoptics Co., Ltd

Osan-si, Gyeonggi-do, South Korea

Engineer

March 2024 - May 2025

Semiconductor Department, R&D Division

- Development of Advanced Glass Singulation Processing Technology
- Development of Next-Generation ABF Drilling Core Technology
- 2024 Advanced Technology Development Project

EDUCATION

Gachon University

Seongnam-si, Gyeonggi-do, South Korea

Undergraduate Student

Expected (February 2026)

College of Semiconductor, Department of Semiconductor Engineering,

Major in Semiconductor Display

Seoul National University

Gwanak-gu, Seoul, South Korea

Exchange Student

August 2024 - February 2025

College of Engineering, Department of Materials Science and Engineering

LEADERSHIP & ACTIVITY

The General Student Club Union

President

June 2025 - Present

- Promoted autonomous and democratic collaboration among scholarly student organizations by leading the General Association of Academic and Research Clubs.

- Facilitated institutional dialogue to safeguard club autonomy and secure university-level support for student-led academic initiatives.
- Built sustainable communication and networking systems to connect student researchers across disciplines and enhance collaborative knowledge exchange.

The Gachon Research Society

Honorary Member

February 2025 - Ongoing

Chairperson of the Acting Executive & Steering Committee for the 2025 Delegates Joint Meeting

February 2025 - Present

Executive Adviser

December 2024 - February 2025

Founding President & 1st President

February 2024 - November 2024

- Contribute to improving language proficiency and fostering academic development through reading and debating multilingual papers and books.
- Facilitate information exchange within the university and strengthen networking opportunities among student researchers.

Student Association, Semiconductor Display Major

Student Association Executive

March 2023 - February 2024

Department of Records and Information

- Conduct research on the given topic or required information, and systematically organize and provide the results.

PUBLICATIONS

- [3] **Hwang, Jeonggyu**. "Estimation of Solar Spectral Irradiance Using Meteorological Data and Analysis of Optimal Conditions for Solar Power Generation." *Under review in Renewable Energy*. Preprint available at arXiv:2504.08008 [eess.SP], 2025.
- [2] **Hwang, Jeonggyu**. "Theoretical Analysis of Doping Concentration Gradients on Solar Cell Performance." *Under review in Next Energy*. Preprint available at arXiv:2407.12245 [physics.app-ph], 2024.
- [1] **Hwang, Jeonggyu**, and Jeon, Jinho. "Exploring the Theoretical Limits of Efficiency in Multilayer Solar Cells." *Submitted to International Journal of Materials Research*. Presented at the 32nd Korean Conference on Semiconductors (KCS 2025). Preprint available at arXiv:2404.14930 [physics.app-ph], 2024.

AWARDS AND HONORS

Academic Excellence Scholarship from Gachon University

Undergraduate Poster Presentation Award

from the 32nd Korean Conference on Semiconductor (KCS 2025)